BOARD OF SECONDARY EDUCATION, KARACHI

SECONDARY SCHOOL (10TH CLASS) EXAMINATION, 2024

CHEMISTRY

Past Paper 2024

Retyped faithfully from the original board paper — same questions, no changes.

SECTION "A" MULTIPLE CHOICE QUESTIONS (MCQs)

Maximum Marks: 12

Note: NOTE: (i) Attempt all the Questions of this Section. (ii) Do not copy down the part questions. Write only the answer against the proper number of the Question and its part according to the question Paper. (iii) Each question carries 1 Marks

Q1. Choose the correct Answer for each from the given question:

Q1. Lactose enzymes are used for:

(A) in production of bread
(B) to make ice cream sweeter
(C) to make Alcohol
(D) To maintain colour

Q2. The reversible chemical reaction will be stable in equilibrium, when:

(A) Qc = Kc
(B) Qc > Kc
(C) Qc < K
(D) None of these

Q3. Insulin which enables our body to use glucose obtained from food or glycogen is basically a:

(A) hormone
(B) enzyme
(C) mineral
(D) starch

Q4. Vital force theory was introduced by:

(A) Berzelius
(B) Wohler
(C) Lewis
(D) Robert Boyle

Q5. Cellulose content in Cotton is:

(A) 70%
(B) 85%
(C) 90%
(D) 95%

Q6. The general formula for Alkyl group is:

(A) C_nH_{2n} (B) C_nH_{2n+1}

(C) C_nH_{2n+2} (D) C_nH_{2n-2}

Q7. Spectroscopy is in Interaction of light with:

(A) Liquid (B) Solid

(C) Gas (D) Matter

Q8. The density of water is maximum at:

(A) −4°C (B) 0°C

(C) 4°C (D) 100°C

Q9. Flame test of Copper Halide with bluish green colour identify the presence of:

(A) Halogen (B) Hydrogen

(C) Copper (D) Nickel

Q10. It is used for surfacing roads:

(A) Neptha (B) Paraffin

(C) Coal (D) Bitumen

Q11. Infrared radiations generated by the earth surface is absorbed in air by higher concentration of:

(A) CH_4 (B) O_3

(C) CO_2 (D) N_2O

Q12. It is not a water borne disease:

(A) hepatitis (B) typhoid

(C) dysentery (D) anemia

SECTION “B” (Short Answer Question)

Maximum Marks: 24

Note: NOTE: Answer any EIGHT questions from this section. Each question carries 3 marks.

Q2. Define

(i) Organic Chemistry

(ii) Bio Chemistry

(iii) Environmental Chemistry

Q3. Describe briefly two types of Buffer Solutions with examples.

Q4. Elaborate the Arrhenius concept of acid and base with suitable examples.

Q5. Frame structural formula of the following Hydro-Carbons:

(i) 2,3 - dimethyl pentane

(ii) 3 – ethyl, 1 – hexene

(iii) 2 – butyne

Q6. What is the atmosphere? List its five major layers.

Q7. State vital force theory and how was this theory discarded by Friedrich Wohler?

Q8. Define:

(i) Titration

(ii) Titrant

(iii) Analyte

Q9. Write one use of the following Enzyme:

(i) Yeast

(ii) Cellulose

(iii) Amyloze

Q10. How vegetable oil is converted into standard fat (ghee)?

Q11. Write down any three adverse impact of Global Warming.

Q12. A solution of HCl has pH of 2.3 Calculate its pOH and $[H^+]$.

Q13. Justify.

(i) Why the coolness on top of a mountain is greater than at sea level?

(ii) Water is considered a universal solvent. Why?

Section C (Detailed –Answer Question)

Maximum Marks: 24

> Note: NOTE: Attempt any FOUR questions from this Section. Each question tries (Six) 6 marks.

Q14. State the law of mass action, apply it on the following reversible reaction. aA + bB ⇌ cC + dD and derive the expression for equilibrium constant.

Q15. What do you mean by quantative analysis? Also define the following terms:

(i) error

(ii) systematic error

(iii) random error

(iv) accuracy

(v) precision

Q16. What are salts? Write down its preparation by action of acid on:

(i) metal hydroxide

(ii) metal carbonate

(iii) metal oxide with representative chemical equation for each.

Q17. What are amino acids? Describe the formation of dipeptide through peptide linkage bond between two amino acid units with structural equation.

Q18. What are carbohydrates? Classify carbohydrates on the basis of hydrolysis into three types with an example of each also mention the source from where these can be obtained.

Q19. Describe how does Ozone depletion occur in stratosphere by chlorofluoro carbons (CFCs) also indicate any three harmful impacts of Ozone depletion.